RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

<table>
<tr><td>PS No.</td><td>41</td></tr>
<tr><td>Project Title</td><td>Restaurant Billing System</td></tr>
<tr><td>Problem Statement</td><td>Write a console billing program for a restaurant where the cashier enters the customer's items and quantities, and the program prints a bill with subtotal, discount, GST and grand total.
Language: Python 3 Concepts: Dictionaries, Loops, Functions, Calculations
Note: Menu: Dosa Rs.80, Idli Rs.40, Thali Rs.150, Tea Rs.20. A 10% discount is given if the subtotal is more than 500. GST is 5% of the amount after discount.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>Order item: Dosa
Quantity: 2</td><td>Dosa x2 added | Amount: 160</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>Order item: Tea
Quantity: 1
Generate bill</td><td>Subtotal: 20 | Discount: 0.00
GST (5%): 1.00 | Grand Total: 21.00</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>(a) Order item: Pizza
(b) Quantity: 0
(c) Quantity: abc</td><td>(a) Item not on the menu. Available: Dosa, Idli, Thali, Tea
(b) Quantity must be at least 1.
(c) Invalid quantity. Enter a whole number.</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>Order Thali x4, Dosa x2, Tea x2
Generate bill</td><td>Thali x4 = 600 | Dosa x2 = 160 | Tea x2 = 40
Subtotal: 800 | Discount (10%): 80.00
GST (5%): 36.00
Grand Total: 756.00</td></tr>
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<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Keep the menu in a dictionary and the ordered items in a list.
3. Use a loop to take many items and write functions to calculate the subtotal, discount, GST and total.
4. Apply the discount rule and the GST rule correctly.
5. Validate the item name and the quantity, and print the bill in a neat formatted layout.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
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